From text to tokens — the fold, then the split

The query and the drawer take exactly this path. Stored bytes are never folded: the promise is verbatim, so this produces comparison keys only.

1 · the fold `normalize::search_key`

one direction, never reversed

- **S0 ascii path** — lowercase, done
- **S1 NFC** — compose
- **S2 compat** — scoped, not NFKC
- **S3 recompose** — if S2 fired
- **S4 lowercase** — must be here
- **S5 strip marks** — + invisibles
- **S6 letter map** — ß→ss, ø→o

Why S4 precedes S5

İ is not a mark, so S5 cannot see it. Lowercasing is what manufactures the U+0307 that S5 then removes.

The observable

`İZMİR → izmir` not the fragment zmi

No Turkish special-casing anywhere in the fold.

Conflations, each pinned by test

`كتابة/كتابه · على/علي · Masse/Maße · πότε/ποτέ · все/всё`

Deliberately NOT stripped

ZWSP is Khmer's word delimiter · ZWJ is contrastive in Malayalam · a blanket Cyrillic strip would make й into и

2 · the split `script::segment`

the script decides, and the script alone

Splitting on “not alphanumeric” finds no boundary at all in Han, Kana, Hangul, Arabic, Khmer, Thai, Lao or Myanmar — a whole clause became one token.

delimiting

Latin · Cyrillic · Greek · Georgian · Tibetan · Devanagari · everything else

whole words

Untouched by segmentation — their defects are folding and morphology, which n-grams do not address.

A joining mark (virama, nukta) is word-INTERNAL here: नमस्ते stays whole.

attaches without a delimiter

Arabic · Hebrew · Khmer · Thai · Lao · Myanmar · Kana · Hangul · Bopomofo

bigrams over same-script runs

Arabic and Hebrew space their words, but their clitics do not: كتاب cannot reach الكتاب by any split.

flagged ngram ⇒ refused the exact slot
Whole-word containment carries it instead.

logographic

Han, and Han only

unigrams + bigrams

A character is a morpheme, so emitting one alone is evidence rather than noise.

NOT flagged as an n-gram — this is the one script where a short unit is a word.

Latin and digit runs inside CJK stay whole, so a brand name survives.

Why the exact slot is refused to n-grams, measured

Bigram-to-bigram equality admitted 74.3% of a real Arabic corpus on one query, against 6.9% for Greek through the same code.

The failure this closed was silence, not bad ranking

Search drops a hit with no lexical evidence and a neutral cosine. Before segmentation, a query for a word the drawer visibly contained shared no token with it at all — so the observable was an empty result set, not a result in the wrong order.